

SE4450 - Final Report Team Ctrl + Incision

1. Executive Summary

This project addresses the challenge of enabling accessible and accurate 3D ultrasound reconstruction for image-guided surgical applications. Traditional ultrasound produces 2D image slices, which limit spatial awareness and require users to mentally reconstruct anatomical structures. While existing solutions such as 3D Slicer provide advanced functionality, they are often complex, resource-intensive, and difficult for students and researchers to customize for focused workflows. This limitation not only increases cognitive load for users but also introduces potential inaccuracies in interpretation, particularly in time-sensitive or precision-critical applications. As a result, improving spatial understanding in ultrasound imaging remains an important challenge in both clinical and research settings.

The objective of this project was to design and implement a modular, student-accessible system that integrates ultrasound data acquisition, spatial tracking, calibration, and 3D reconstruction into a streamlined pipeline. The system combines hardware components, including an ultrasound probe, frame grabber, and NDI Aurora electromagnetic tracker, with software tools such as Plus Toolkit and a Python-based interface to enable tracked freehand ultrasound imaging.

The developed system successfully captures synchronized ultrasound and tracking data, generates MetalImage (.mha) files with embedded spatial metadata, and reconstructs these sequences into 3D volumetric representations. A calibration subsystem, including intrinsic camera calibration, pivot calibration, and hand-eye calibration, ensures spatial accuracy throughout the pipeline. The integration of a PySide6-based graphical interface further improves usability by allowing users to configure hardware, perform calibration, and visualize reconstruction results within a unified environment. Additionally, the system was designed to preserve both image data and spatial metadata throughout the pipeline, ensuring that each captured frame could be accurately positioned within a global coordinate system during reconstruction.

The project achieved its primary goal of demonstrating a functional end-to-end pipeline for tracked ultrasound acquisition and 3D reconstruction using open-source tools. However, several limitations remain, including sensitivity to calibration accuracy, reliance on manual configuration steps, and incomplete automation of the reconstruction workflow. These challenges highlight areas for future work, such as improving robustness, reducing user overhead, and enhancing system integration. Furthermore, the system's performance is highly dependent on the quality of input data, including the consistency of ultrasound acquisition and the accuracy of calibration procedures. Variability in these factors can significantly impact reconstruction quality, emphasizing the need for improved validation mechanisms and user guidance.

Overall, this project contributes a lightweight and customizable alternative to existing medical imaging platforms, lowering the barrier to entry for students and researchers working in image-guided intervention and medical imaging. Despite these limitations, the system demonstrates the feasibility of building accessible, research-oriented medical imaging tools using open-source technologies and highlights the potential for further development in this space.

2. Introduction & Background

Image-guided surgical systems integrate medical imaging, spatial tracking, and visualization technologies to provide accurate representations of anatomical structures. Ultrasound imaging is widely used in this domain due to its low cost, portability, and real-time capabilities. However, conventional ultrasound produces a series of independent two-dimensional image slices, requiring users to mentally reconstruct three-dimensional structures. This lack of inherent spatial context introduces challenges in interpretation and limits the effectiveness of ultrasound in applications requiring precise spatial awareness.

To address this limitation, 3D ultrasound reconstruction techniques have been developed to transform sequential 2D images into volumetric representations. Freehand ultrasound systems, which combine tracked probe motion with image acquisitions, enable the reconstruction of 3D volumes by associating each image frame with spatial position and orientation data. While this approach improves anatomical visualization and supports more accurate diagnostics and interventions, existing implementations are often difficult to use and require significant technical expertise.

Current software platforms, such as 3D Slicer and the Plus Toolkit, provide powerful capabilities for medical image processing and reconstruction. However, these systems are typically designed as comprehensive, all-in-one solutions that introduce substantial complexity and steep learning curves. Their architecture, often built in low-level languages such as C++, can limit accessibility and flexibility for students and researchers who require lightweight, customizable tools for specific tasks such as ultrasound data acquisition and reconstruction. As a result, there exists a gap between the capabilities of existing systems and the needs of users seeking modular and accessible solutions.

The primary stakeholders of this project include students and researchers working in medical imaging and image-guided intervention, particularly those in academic environments such as Robarts Research Institute. These users require tools that support experimentation, learning, and rapid development without the overhead of complex clinical software systems. Secondary stakeholders include developers who may extend the system and, more broadly, clinicians who could benefit from simplified and customizable imaging workflows in non-clinical or research settings.

The motivation for this project is to bridge this accessibility gap by developing a modular, lightweight system for tracked ultrasound acquisition and 3D reconstruction. The goal is to provide a streamlined alternative that reduces software bloat while maintaining core

functionality, enabling users to interact with hardware at a lower level and adapt the system to different use cases. By focusing on usability, modularity, and integration, the project aims to support both educational and research applications in medical imaging.

The development of this system was subject to several constraints. These included the complexity of integrating hardware components such as the NDI Aurora electromagnetic tracker, ultrasound probe, and frame grabber, as well as the limited documentation and configuration challenges associated with tools such as Plus Toolkit. Additionally, the project was conducted within the time constraints of an academic capstone, requiring trade-offs between feature completeness, system integration, and validation. Despite these challenges, the project aimed to deliver a functional and extensible system that demonstrates the feasibility of accessible 3D ultrasound reconstruction using open-source technologies.

3. Requirements

The project was designed as a modular system that supports calibration, tracked data acquisition, reconstruction, and visualization. The requirements outlined below reflect both the original project goals and the scope of the final implemented system.

Functional Requirements

The functional requirements describe the core behaviors and capabilities the system was expected to provide.

The system must support the acquisition of tracked ultrasound image data by integrating an ultrasound image stream with spatial tracking information from the NDI Aurora tracker. This functionality includes capturing synchronized image and position/orientation data during a recording session to ensure accurate tracking.

A complete calibration workflow is provided for the tracked stylus and camera subsystem. This includes camera intrinsic calibration, pivot calibration, axial calibration, hand-eye calibration, and projected stylus tip verification. The calibration process establishes the geometric relationships required for accurate tracking and visual overlay.

The system must generate tracked ultrasound sequence files, primarily in MetaImage format (.mha), containing both image data and the spatial metadata required for reconstruction.

Additionally, offline 3D volume reconstruction from tracked ultrasound sequence files is supported, allowing users to select the appropriate reconstruction configuration and output location.

The system must provide a viewer capable of loading and rendering reconstructed 3D volumes. The viewer needs to support interactive operations such as zooming, rotating, and translating the rendered volume.

A simplified user interface is also provided to guide users through the workflow of calibration, hardware configuration, acquisition, reconstruction, and viewing, without requiring the use of command-line tools for standard use cases.

The system should have a lightweight database backend to maintain local records of capture sessions, device configurations, tracking streams, and logs.

Non-Functional Requirements

The non-functional requirements define the system's key characteristics and engineering constraints, including modularity, usability, availability, accuracy, and extensibility.

The system is structured as a modular pipeline, with calibration and acquisition, reconstruction, and visualization into separate components. This allows individual subsystems to be tested, modified, and debugged independently.

The user interface is designed to provide a lightweight and focused experience, reducing the complexity associated with larger medical imaging platforms. It should support common tasks with minimal setup burden for users unfamiliar with advanced medical imaging workflows.

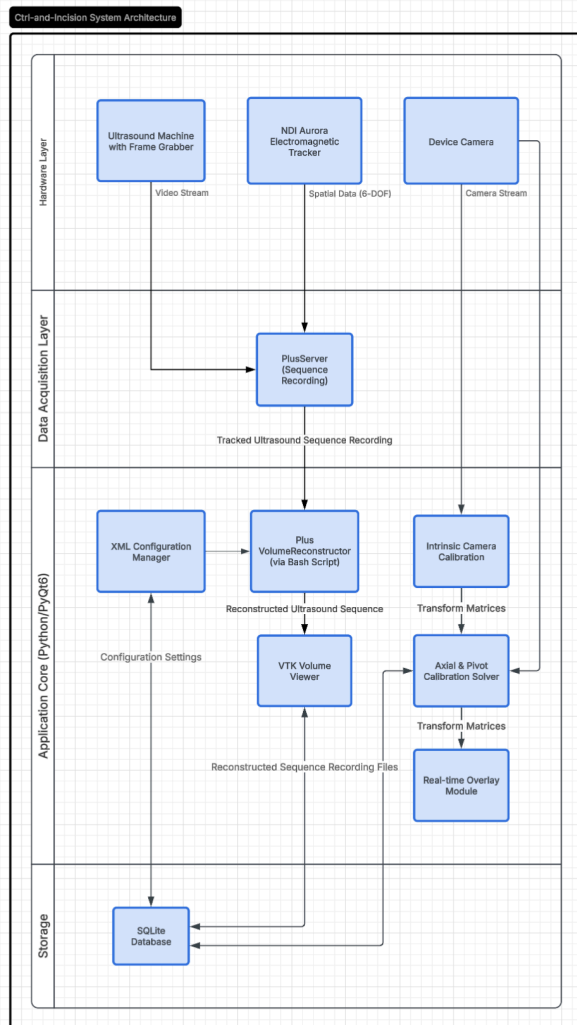
The system's core functions remain consistently accessible through a single workflow without excessive manual setup or tool switching.

The system maintains spatial consistency between tracked stylus, ultrasound data, and reconstructed output. Calibration and transform accuracy are required to achieve valid reconstruction and tracking performance.

Finally, the system architecture supports the integration of future modules without requiring a complete system redesign.

4. Architecture & Design

System Architecture Design



Key Design Decisions and Justifications

The design of Ctrl-and-Incision was guided by the need to move away from "black-box" medical software toward a transparent, research-oriented platform.

- **Modular Architecture:** We chose to build a custom modular system rather than using existing platforms such as 3D Slicer. This decision allows for a lightweight application footprint and provides full developer control over the UI and data flow, which is essential for specialized surgical research.
- **Decoupled Acquisition and Reconstruction:** The system separates the real-time data acquisition (PlusServer) from the reconstruction and visualization stages. This

justification is based on the need to test each component independently, reducing system-wide failures when hardware synchronization issues occur.

- **Clinical UI Refactoring:** We decided to use a simplified interface that moves away from overly technical language and ensures ease of use through clear and intuitive workflows. This was justified by user feedback indicating that the UI in existing applications are too complex for a surgical environment.
- **Simulation-Ready Development Pipeline:** A critical decision was to implement a simulation-ready pipeline using a Boolean flag. This allowed the team to continue software development and testing of the reconstruction logic even when physical ultrasound hardware or NDI trackers were unavailable.
- **Transformation Chain Transparency:** We chose to explicitly implement and document the coordinate transformation pipeline (Image -> Probe -> Tracker -> World). This ensures the high spatial accuracy required for volume reconstruction, which was a primary concern raised by project advisors.

Technology Choices and Trade-offs

Technology	Role	Trade-offs and Reasoning
Plus Toolkit	Data Acquisition & Synchronization	Reasoning: It is the industry standard for hardware-independent data acquisition and synchronization. Trade-off: It introduces a significant integration burden due to complex XML configuration management and manual setup requirements.
VTK	3D Volume Rendering	Reasoning: Provides precise volumetric rendering and control over scalar opacity and colour presets. Trade-off: VTK has a steep learning curve and requires intensive management of the rendering loop to maintain performance.
NDI Aurora	Electromagnetic Tracking	Reasoning: Offers high-precision 6-DOF tracking without the line-of-sight requirements of optical systems. Trade-off: Requires careful temporal calibration to ensure the tracking data aligns perfectly with ultrasound frames.
Python & PyQt6	Application Framework	Reasoning: Allows for rapid prototyping and easy integration of various sub-processes (like bash scripts for reconstruction through Plus Toolkit). Trade-off: Managing Python dependencies across different laboratory machines can lead to environment mismatches.
ArUco Markers	Stylus Tracking	Reasoning: Enables low-cost, immediate visual validation of pose estimation correctness. Trade-off:

		Accuracy is highly dependent on camera calibration and lighting conditions.
--	--	---

5. Implementation

The system was designed as a modular, student-accessible application for image-guided surgery, organized into four primary components: hardware configuration, data acquisition, 3D reconstruction, and visualization. Alongside these, the system incorporated a calibration subsystem consisting of camera intrinsic calibration, pivot calibration, and ArUco marker tracking. Each module was developed to function both independently and as part of the broader pipeline, with the goal of producing a cohesive tool that could be used and extended by students without deep familiarity with the underlying hardware or medical imaging infrastructure.

Hardware configuration was one of the most technically demanding aspects of the project. Integrating the Aurora NDI electromagnetic tracker, the ultrasound device, and a frame grabber required each component to be recognized and correctly configured within the Plus Server before any data could flow through the pipeline. This process involved significant trial and error, and the team ultimately consulted master's students in the lab to resolve configuration issues that were not well-documented. A key outcome of this phase was obtaining the image-to-probe spatial transform for the ultrasound transducer, calibrated specifically for imaging depths of 8mm and 9mm with the Aurora NDI system. This transform was essential for accurately mapping 2D ultrasound frames into 3D space.

Data acquisition was managed through Plus Server, which acted as the middleware layer between the physical hardware and the rest of the software pipeline. The server streamed tracked ultrasound frames and packaged them into .mha metaimage files, a format that encodes both image data and associated spatial metadata. These files served as the input to the reconstruction stage, allowing the pipeline to be decoupled such that acquisition and reconstruction could be tested and developed somewhat independently.

The .mha file format was the primary data structure used to pass imaging data between pipeline stages. An .mha file stores image data alongside a header containing critical spatial metadata, including the size and spacing of the image, the coordinate transform associated with the frame, and the data type of the pixel values. In the context of this project, each .mha file captured by Plus Server encoded not just the raw ultrasound image but also the spatial position and orientation of the probe at the moment of capture, provided by the Aurora NDI tracker. This tight coupling of image and transform data within a single file is what makes the format particularly well-suited to tracked medical imaging, as it allows the reconstruction stage to correctly place each 2D frame in 3D space without requiring a separate synchronization mechanism.

3D reconstruction was handled using VolumeReconstructor.exe, a tool within the Plus Server ecosystem designed to take a sequence of tracked 2D ultrasound frames and compose them into a coherent 3D volume. A significant challenge during this phase was determining the correct reconstruction settings, parameters governing how frames were interpolated and combined had a substantial effect on output quality, and finding a configuration that produced accurate and usable volumes required iterative testing. Once a volume was generated, it was passed to a VTK-based rendering pipeline for display. Applying appropriate transfer functions to distinguish tissues and structures of varying density proved to be another challenge, as the mapping between raw intensity values and meaningful visual output required careful tuning to produce clinically interpretable results.

The scikit-surgery library provided a high-level Python interface for interacting with surgical tracking hardware and was used in this project primarily to facilitate communication with the Aurora NDI electromagnetic tracking system. scikit-surgery abstracts much of the low-level complexity involved in querying tracking devices, offering consistent data structures for retrieving tool positions and orientations in real time. This allowed the team to integrate NDI tracking data into the Python-based pipeline without having to interface directly with the device's native SDK, significantly reducing implementation complexity. The library is part of a broader ecosystem developed for surgical navigation research, and its use here reflects the project's alignment with real clinical and academic workflows in image-guided intervention.

The calibration subsystem served to ensure spatial accuracy throughout the pipeline. Intrinsic camera calibration characterized the optical properties of the imaging system, while point calibration and ArUco marker tracking were used to establish and verify spatial relationships between coordinate frames. Together, these calibration steps ensured that the 2D-to-3D reconstruction was grounded in accurate geometric transforms, which is critical in an image-guided surgery context where spatial precision directly affects usability.

ArUco marker tracking was implemented as part of the camera calibration module, providing a method for tracking a physical probe within camera space. ArUco markers are square fiducial markers with a unique binary pattern that can be reliably detected and identified by a camera in real time. Using OpenCV's ArUco detection pipeline, the system was able to identify marker positions within the camera's field of view and compute the corresponding pose, which is the position and orientation of the tracked probe relative to the camera. While this module operates independently from the ultrasound-to-3D reconstruction pipeline, it was included to give the system completeness as an image-guided surgery platform, ensuring that camera-based tracking and spatial calibration were available alongside the electromagnetic tracking provided by the Aurora NDI.

The frontend was built using PySide6 and Qt, providing a unified graphical interface that served as the control panel for the entire application. A single landing page gave users access to both the calibration module and the ultrasound-to-3D imaging pipeline, keeping the interface approachable for student users unfamiliar with the underlying system complexity. The scikit-surgery library was also incorporated to support surgical tool tracking and related functionality. Integrating the frontend with the backend pipeline required careful coordination to

ensure that user actions in the UI correctly triggered the appropriate processing steps, and that reconstruction and visualization outputs were displayed responsively within the application.

6. Verification & Testing

Verification and testing were performed throughout development to confirm that both individual modules and the full workflow behaved as expected.

Testing was first carried out at the component level to verify that individual subsystems functioned correctly independently. Calibration and the stylus tracking subsystem was also tested independently by verifying ArUco marker detection, live stylus tracking, and projected overlay behavior in the camera feed. In addition, the 3D viewer was tested to confirm that reconstructed .mha files could be loaded and rendered correctly, and that users could interact with the rendered volume through the standard interactive operations. The SQLite backend was also verified through session creation, capture session storage, and retrieval of recent session records.

Integration testing was then used to verify interaction between the key system components, because many project challenges resulted from inconsistencies between configuration files, transform definitions, hardware assumptions, and metadata. This level of testing focused on confirming correct interaction between the Python interface and PLUS Toolkit executables, proper loading and modification of PLUS XML configuration files, successful communication between camera-based stylus tracking and the live video pipeline, integration of NDI Aurora tracking data with ultrasound capture, and successful generation of tracked .mha sequence files containing both image data and spatial metadata.

At the highest level, system testing was performed on the full end-to-end workflow. The workflow under test was the complete sequence of configuring the hardware, recording tracked ultrasound data, generating a tracked sequence file, reconstructing that file into a 3D volume, and visualizing the result in the viewer. Successful execution of this workflow demonstrated that the system could function as a complete working pipeline.

Additionally, the system was repeatedly tested from the user's perspective to ensure that the intended workflow could be completed without requiring manual and low-level interventions.

Overall, testing covered the major implemented capabilities of the system. Verified areas included calibration workflow components, ArUco marker detection and stylus tracking, XML configuration editing and usage, tracked sequence generation, PLUS Toolkit reconstruction execution, 3D volume loading and rendering, database session and stream storage, and the full acquisition-to-reconstruction workflow using sample data. Some areas were only partially verified, including reconstruction consistency across different scan qualities and transform reliability under varied setup conditions. Other areas were not fully verified, such as quantitative spatial accuracy and system stability across a wider range of scan targets or hardware configurations.

Validation of the reconstructed outputs was performed primarily through qualitative inspection and end-to-end workflow testing. The system was tested on representative datasets to confirm that tracked sequence generation and reconstruction behaved correctly under controlled conditions. It was also tested using real captured data, with the reconstructed volumes inspected to ensure they were continuous, structurally plausible, and free of major geometric errors. In addition, reconstructed .mha outputs were loaded in 3D Slicer and compared against the system's own viewer. This provided useful confirmation that the outputs were structurally valid and compatible.

Testing also revealed several important known defects and limitations. Reconstruction quality was found to depend heavily on scan technique. For example, moving the probe too quickly, lifting it away from the phantom, or sweeping inconsistently could produce gaps, distortions, or missing regions in the final volume. The system was also highly sensitive to transform correctness, meaning that even when image data was captured successfully, small errors in calibration or alignment could lead to poor spatial placement of slices and result in degraded reconstruction quality. Additional issues were observed when crop settings were not well constrained, which could introduce unwanted image regions such as text overlays, air, or irrelevant borders into the reconstruction. Finally, while the stylus pivot and axial calibration workflow was functional, it is not fully automated.

Despite these limitations, the testing performed was sufficient to verify the project as a working prototype. The results confirmed the primary engineering objective, a functional end-to-end pipeline for tracked ultrasound acquisition, calibration, reconstruction, and visualization, was successfully achieved. The testing also identified factors affecting system performance, providing clear direction for future work.

7. Results & Discussion

The primary success criterion for Ctrl + Incision was the completion of a full end-to-end pipeline, from live ultrasound acquisition through to 3D volume reconstruction and visualization. This criterion was met: the system successfully captures tracked ultrasound frames via the Aurora NDI electromagnetic tracker and framegrabber, streams them through Plus Server as .mha files, and reconstructs a 3D volume rendered in the PySide6/Qt-based interface using VTK.

During demonstration and testing, the reconstructed volumes were recognizable and structurally coherent, confirming that the core pipeline, hardware integration, spatial tracking, frame acquisition, and volume reconstruction, functions as intended. However, several issues were identified through team testing and external feedback that affected the quality of the final reconstruction.

First, the raw ultrasound frames were not cropped to isolate the anatomical data from the fan-shaped sector display. As a result, the reconstruction included non-diagnostic regions of each frame, introducing visual noise into the rendered volume. Implementing a masking or

cropping step to extract only the active sector region before reconstruction would address this in a future iteration.

Second, scanning speed proved difficult to calibrate. When the probe was moved too quickly across the surface, the spatial gaps between consecutively captured frames became large enough that the reconstruction algorithm could not interpolate smoothly between them, producing visible gaps in the volume. This points to a need for either real-time feedback to the operator indicating scan pace, or a denser frame capture rate to tolerate faster motion.

Third, lifting the probe away from the scanning surface during acquisition introduced floating artifacts; disconnected clusters of voxel data spatially offset from the main volume. These artifacts resulted from tracked frames being captured while the probe was not in consistent contact with the target, causing the reconstruction to place data at incorrect spatial positions. A contact-detection mechanism or post-processing filter to discard outlier frames could mitigate this issue.

Despite these quality issues, the project met its defined success criterion of a complete, functioning pipeline. The errors identified are refinement-level concerns rather than fundamental architectural failures, and each has a clear path to resolution. The system demonstrates that low-cost, real-time 3D ultrasound reconstruction using open-source tools (Plus Server, VTK, OpenCV) is feasible, and provides a solid foundation for future development toward clinical applicability.

8. Conclusions & Future Work

The project successfully delivered a complete end-to-end pipeline encompassing ultrasound frame acquisition via a framegrabber, electromagnetic spatial tracking through the Aurora NDI system, data streaming and storage through Plus Server as .mha files, 3D volume reconstruction using VTK, and an interactive visualization interface built with PySide6/Qt. Additionally, the system incorporated ArUco marker tracking via OpenCV and instrument visualization through scikit-surgery, laying groundwork for tool-to-anatomy spatial registration. The completion of this full pipeline represents the project's core accomplishment and satisfies the primary success criterion established at the outset.

That said, an honest assessment of the prototype reveals meaningful shortcomings. The absence of an ultrasound frame cropping step meant that the fan-shaped sector boundary was included in reconstructions, introducing visual noise that would be undesirable in a clinical context. Scan quality proved highly sensitive to operator technique, moving the probe too quickly produced gaps in the volume, and lifting the probe generated floating artifacts of disconnected voxel data. The team also underestimated the learning curve associated with hardware integration and real-time data streaming, which consumed a significant portion of the project timeline and limited the amount of iteration possible on reconstruction quality.

Several specific enhancements are recommended for future work. The most immediate priority is implementing a frame preprocessing step that masks or crops each ultrasound frame to isolate the active sector region before it enters the reconstruction pipeline. This is a relatively straightforward image processing task and would significantly improve volume clarity. Second, a contact detection or outlier rejection filter should be developed to identify and discard frames captured while the probe is not in stable contact with the scanning surface, eliminating the floating artifact problem at its source. Beyond these corrections, longer-term enhancements could include integrating deformable registration to account for tissue motion between frames, supporting multiple imaging modalities, and conducting formal usability testing with clinicians to evaluate the interface under realistic surgical planning scenarios.

Overall, Ctrl + Incision demonstrates that a low-cost, open-source 3D ultrasound reconstruction system is technically achievable using accessible tools and hardware. While the prototype is not yet suitable for clinical deployment, it provides a solid architectural foundation and identifies clear, actionable directions for refinement. The project represents a meaningful step toward making image-guided surgical planning more accessible and affordable.

9. References

- [1] SciKit-Surgery, "VTK Overlay With ArUco Tag," [scikit-surgerytutorial01.readthedocs.io](https://scikit-surgerytutorial01.readthedocs.io/en/latest/03_VTK_Overlay_With_Aruco_Tag.html). Available: https://scikit-surgerytutorial01.readthedocs.io/en/latest/03_VTK_Overlay_With_Aruco_Tag.html
- [2] Kitware, "VTK Documentation," [vtk.org](https://vtk.org/documentation/). Available: <https://vtk.org/documentation/>
- [3] Insight Software Consortium, "MetaIO Documentation," ITK Wiki Archive. Available: <https://insightsoftwareconsortium.github.io/ITKWikiArchive/Wiki/ITK/MetaIO/Documentation/>
- [4] PlusLib, "Volume Reconstructor Application," [pluslib.readthedocs.io](https://pluslib.readthedocs.io/en/latest/applications/ApplicationVolumeReconstructor.html). Available: <https://pluslib.readthedocs.io/en/latest/applications/ApplicationVolumeReconstructor.html>
- [5] Northern Digital Inc., "Aurora Electromagnetic Tracking System," [ndigital.com](https://www.ndigital.com/electromagnetic-tracking-technology/aurora/). Available: <https://www.ndigital.com/electromagnetic-tracking-technology/aurora/>